Tokens-per-Parameter Coverage Is Critical for Robust LLM Scaling Law Extrapolation

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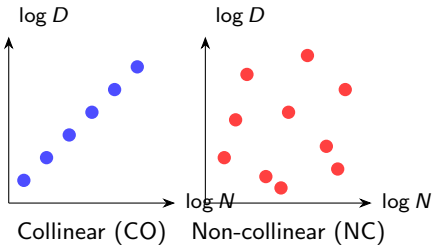






Collinear designs

- Scaling laws are fitted on small runs and used to predict large ones.
- Many studies train at a fixed tokens-per-parameter ratio $k = D/N$: a ray.
- On a ray both power-law terms move together:
  $L(N, kN) = AN^{-\alpha} + Bk^{-\beta}N^{-\beta} + E.$
- Only the exponent gap $\varepsilon = |\alpha - \beta|$ separates A from B.



Theory: when the scale pair is identified

- One ray: $\kappa_{A,B} = \Theta(\varepsilon^{-2})$, and the interval on A is $\Omega(\varepsilon^{-1})$ times the one on the combined $\psi = A + Bk^{-\alpha}$.
- K rays $k_1 < \cdots < k_K$: to leading order $\kappa_{A,B} \le \kappa_{\mathrm{target}}$ whenever

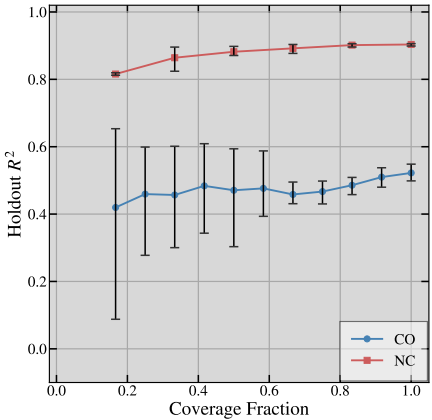
$$\underbrace{\frac{1}{K}\sum_{\ell=1}^{K} k_\ell^{-2\beta_{\mathrm{eff}}} - \Big(\frac{1}{K}\sum_{\ell=1}^{K} k_\ell^{-\beta_{\mathrm{eff}}}\Big)^2}_{V_K} \;\ge\; \underbrace{\frac{\big(K + \sum_\ell k_\ell^{-2\beta_{\mathrm{eff}}}\big)^2}{K^2\,\kappa_{\mathrm{target}}}}_{\tau_K},$$

- and $\kappa_{A,B} \le \kappa_{\mathrm{target}}$ iff $V_K \ge \tau_K/(1 + 1/\kappa_{\mathrm{target}})^2$.
- One ray never passes; two rays pass once $k_2/k_1 \ge R_{\min}(k_1)$, a check made before training.

Results: non-collinear designs extrapolate

Comparison (NC vs. CO)	NC wins
Full fan, FP32 (1,460 of 1,500)	97.3%
Full fan, BF16 (296 of 300)	98.7%
Budget-matched, Chinchilla	68.5%
Budget-matched, Droppo-Elibol	57.4%
Budget-matched, Kaplan	80.6%

- About 1,900 models, 5.04M-76.5M parameters, five corpora, four laws.
- Budget-matched rows: CO subsets below the threshold.



Holdout R^2 against TPP coverage (Kaplan, RedPajama).

Takeaway and design recipe

- Report coefficients only from designs that span TPP ratios.
- Recipe: pick k_1, read $R_{\min}(k_1)$, train two ratios at the ends of that spread, split runs evenly, and check the exact $\kappa_{A,B}$ before training.
- Code, data and checkpoints are public.

`lab-v2.github.io/tpp-coverage`

